

Week 5 Report

This Week in Short

This week had two focuses. The first was sharpening the vision side of the project so the arm sees more reliably. The second was writing a guide book for the xArm 6 that students and trainees can follow on their own.

Sharpening the Vision

Most of the week went into tightening the YOLO detection the arm relies on. The aim was fewer missed cubes, fewer false alarms, and steadier readings of where a cube actually sits, so the arm is not reacting to noise. The detection is noticeably more dependable now, and I am continuing to feed it more of our own captured images so it keeps improving under the lighting and background we really work in.

A Guide Book for the Trainees

Alongside that, I started putting together a guide book for the xArm 6 aimed at students and new trainees. The idea is a single place that covers the basics: how to power up and connect to the arm safely, how to move it and teach it positions, how to run the ready-made buttons we set up, and what to do when it stops or reports an error. It is written plainly, with step-by-step instructions, so someone new can get the arm working without having to be walked through it in person.

Next Week

Carry on refining the vision with more training data, and finish and tidy up the guide book so it can be handed to the trainees.